MASTER REFERENCE STANDARD: COMBINED-CYCLE TURBINE UNIT #4

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1. Executive Summary & Heat Balance Analysis

This report documents the quarterly performance verification of Combined-Cycle Gas Turbine Unit #4 in accordance with international turbine testing standards [1] and plant baseline specs [2]. During full-load steady-state operation, the primary thermodynamic cycle exhibited stable combustor dynamics across all three monitoring zones.

The combined auxiliary thermal load across the three heat recovery loops is calculated as: Total Thermal Load = Loop A + Loop B + Loop C: 1250.0 + 840.0 + 310.0 = 2400.0 kW.

Parasitic electrical consumption over the 6-hour baseline window reached 420.5 kWh, while the exhaust manifold sensor recorded an average plenum temperature of 315 K. Detailed harmonic distortion plots are provided in Appendix A, and the rotor dynamic vibration spectrum is documented in Appendix C and validated against citation [4].

2. Subsystem Power & Energy Allocation Table

Subsystem Module	Rated Power (kW)	Operating Hours	Total Energy (kWh)
HP Feedwater Pump	25.0	24.0	600.0
Lube Oil Circulator	12.5	24.0	300.0
Condenser Vacuum Unit	15.0	20.0	300.0
Cooling Tower Fan Bank	11.25	20.0	225.0
Total	**63.75**	**88.0**	**1425.0**

Note: Subsystem telemetry was cross-checked against historical baseline dataset [3].

3. Generator Excitation & Sensor Calibration

Field winding insulation resistance was verified prior to synchronization using a 5 kV megohmmeter. Following transducer calibration, the nominal bus voltage under 85% power factor load was evaluated using the linear tap scaling formula: 480.0 * 0.85 = 408.0 V across the secondary bus. Acoustic emission thresholds complied with baseline limits defined in Report TR-2025-88B [4]. Thermal expansion clearances for the low-pressure casing are summarized in Appendix B.

4. Overspeed Trip & Safety Interlock Verification

redundant electro-hydraulic trip solenoids actuated within 18 ms at 110% rated rotor speed (3300 RPM).

Appendix A: Harmonic Distortion Summary

Total Harmonic Distortion (THD) remained below 1.8% across all three phases during full-load testing.

Appendix B: Casing Thermal Clearance Log

Axial rotor-to-stator clearance measured 4.2 mm (minimum allowable threshold: 3.0 mm).

Appendix C: Rotor Dynamic Vibration Spectrum

Peak bearing housing vibration amplitude was 1.4 mm/s RMS, well below the ISO 10816 limit of 2.8 mm/s.

References

[1] ASME PTC 46-2015, Performance Test Code on Overall Plant Performance, American Society of Mechanical Engineers.

[2] ISO 2314:2009, Gas turbines - Acceptance tests, International Organization for Standardization.

[3] Plant Engineering Archive, Unit #4 Commissioning & Baseline Heat Rate Log (Document ID: BL-2024-01).

[4] Acoustic & Vibration Laboratory Report TR-2025-88B, Rotor Dynamic & Acoustic Baseline Qualification.